

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

### 1. Asset replacement problem

At the beginning of each year, a manufacturer must decide whether to continue to operate an aging asset or replace it with a new one. An asset that is  $a$  years old (at the beginning of each year) yields a profit contribution  $p(a)$  up to  $n$  years, after which it becomes unsafe and must be replaced. The cost of a new asset is  $c$ .

This is an infinite-horizon, deterministic problem. The state variable is age:  $a = \{1, 2, \dots, n\}$ . The control variable is  $x = \{keep, replace\}$ . The state-transition function is

$$g(a, x) = \begin{cases} a + 1, & x = keep \\ 1, & x = replace \end{cases}$$

The instantaneous payoff function is

$$f(a, x) = \begin{cases} p(a), & x = keep \\ p(0) - c, & x = replace \end{cases}$$

The value of an asset of age  $a$  satisfies the Bellman equation

$$V(a) = \max\{p(a) + \beta V(a + 1), p(0) - c + \beta V(1)\},$$

where we set  $p(n + 1) = -\infty$  to force the manufacturer to replace the asset. Therefore,  $V(n + 1) = p(0) - c + \beta V(1)$ . Note that  $V(a)$  measures the NPV of profits.

For your numerical solution, assume:  $p(a) = 50 - 2.5a - 2.5a^2$ ,  $c = 75$ ,  $\beta = 0.9$ ,  $n = 10$ .

- Solve for the value function  $V(a)$  and plot it. Note that age  $a$  takes a finite number of values, and hence  $V(a)$  is a vector.
- What replacement policy maximizes profits?
- Suppose that in period 1,  $a = 1$ . Plot the path of the state variable (age  $a$ ) given an optimal policy for  $t = 1, \dots, 16$ .